

COL1000: Lab 8

Semester 2, 2025–26

Instructions Upload your code to Gradescope as: `<entryNumber>_q<quesNumber>.py`. After uploading, run the autograder to check that all tests pass. Make sure not to include any custom prompt text in `input()` calls, and any extra text in `print()` statements. You must not use the `math`, `ast`, `intertool`, `sympy` libraries and advanced built-in python functions such as `split`, `splitlines`, `map`, `join`.

Question 1: Composite Numbers Write a program that reads two integers n and k on separate lines. Then outputs a list of the smallest n composite numbers which have at least k distinct prime factors.

Example:

Input:

5
3

Output:

[30, 42, 60, 66, 70]

Question 2: List operations Write a program that first reads two integers n and k on separate lines. Then reads n integers on separate lines and stores them in a list. After that, reads k commands of the following form: (A) **append** x (append x to the end of the list), (B) **insert** i x (insert x at index i), (C) **remove** x (remove the first occurrence of x), (D) **pop** i (remove the i -th element), and performs the corresponding operations. After all operations are performed, it outputs the final list.

Example:

Input:

5
3
15
16
19
22
16
remove 16
insert 1 5
pop 3

Output:

[15, 5, 19, 16]

Question 3: Number of moves Given an $m \times n$ grid of 0's (free cells) and 1's (obstacles), a robot starts at $(0,0)$ and wants to reach $(m-1, n-1)$, moving only right or down and never stepping on a 1. Write a Python program that implements a recursive function to calculate the total number of unique valid paths from the start to the destination.

The first line of input contains three space separated integers: m (rows), n (columns), and k (number of obstacles). The following k lines each contain two integers, r and c , separated by a single space representing the row and column indices of obstacle.

Example:

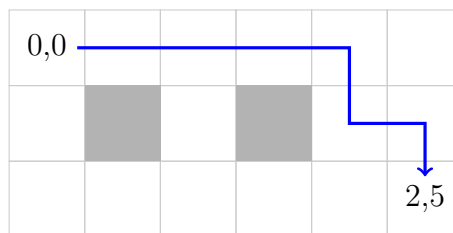
Input:

```
3 6 2
1 1
1 3
```

Output:

5

Explanation: The total number of unique valid paths from $(0,0)$ to $(2,5)$ in the grid below is 5.



Hint: For any cell, the number of paths to it from $(0,0)$ is the sum of the paths from the cell above and the cell to the left.

Question 4: Permutation Write an iterative/recursive code that reads an integer n and prints a sorted list of integers obtained by permuting the digits of n .

For example, for $n=1150$, the output must be:

```
[115, 151, 511, 1015, 1051, 1105, 1150, 1501, 1510, 5011, 5101, 5110]
```

Note that numbers like 150, 105, 510, and 501 are not included, as they do not contain two copies of the digit 1.